

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: DATA ANALYSIS WITH SPSS/SAS/R

PRACTICAL 10

AIM: Creating new variables using transformations and calculations in R. import dataset.

>COMMANDS USED

#

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R Script: Creating New Variables (Transformations & Calculations)

Dataset: AI Job Impact Data

#

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library(dplyr)

library(tidyr)

library(stringr)

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#

1. SETUP: Import the Dataset

#

Import data

ai_jobs_df <-

read.csv("AI_Impact_on_Jobs_2030.csv")

print("--- Baseline Data Sample ---")

**# Show key columns used for the upcoming
calculations**

**print(head(ai_jobs_df, 3) %>% select(Job_Title,
Average_Salary, Years_Experience,
AI_Exposure_Index))**

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2. METHOD A: ARITHMETIC CALCULATIONS

#

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Scenario 1: Calculate Salary Efficiency (Salary per Year of Experience).

Formula: $\text{Average_Salary} / \text{Years_Experience}$

Scenario 2: Create a composite "Future Readiness Score". (Higher score is better)

Formula: $(1 - \text{Automation_Probability_2030}) + (1 - \text{AI_Exposure_Index}) / 2$

`df_calc <- ai_jobs_df %>%`

`mutate(`

`Salary_Per_Year = Average_Salary /`
`Years_Experience,`

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```
Future_Readiness_Score = ( (1 -  
Automation_Probability_2030) + (1 -  
AI_Exposure_Index) ) / 2  
)
```

```
print("--- Method A: Arithmetic Results (Salary  
Efficiency & Readiness Score) ---")  
print(df_calc %>% select(Job_Title, Average_Salary,  
Years_Experience, Salary_Per_Year,  
Future_Readiness_Score))
```

```
#
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```
# 3. METHOD B: CONDITIONAL LOGIC (case_when  
and ifelse)
```

```
#
```

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```

```
# Scenario 1: Create a categorical 'Experience_Level'  
based on Years_Experience.
```

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We use the versatile case_when() function for multiple conditions.

Logic:

Years_Experience <= 5: "Junior"

Years_Experience > 5 AND <= 15: "Mid-Level"

Years_Experience > 15: "Senior/Expert"

```
df_logic <- ai_jobs_df %>%  
  mutate(  
    Experience_Level = case_when(  
      Years_Experience <= 5 ~ "Junior",  
      Years_Experience > 5 & Years_Experience <=  
15 ~ "Mid-Level",  
      Years_Experience > 15 ~ "Senior/Expert",  
      TRUE ~ "Unknown" # Default catch-all  
    ),  
  )
```

Scenario 2: Simple binary check for low automation risk

```
Is_Low_Automation =  
ifelse(Automation_Probability_2030 < 0.20, TRUE,  
FALSE)
```

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)

```
print("--- Method B: Logic Results (Experience Level  
& Low Automation Flag) ---")
```

```
print(df_logic %>% select(Years_Experience,  
Experience_Level, Automation_Probability_2030,  
Is_Low_Automation))
```

```
#
```

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```

```
# 4. METHOD C: TEXT TRANSFORMATION (paste)
```

```
#
```

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```

```
# Scenario: Create a 'Job_Summary' that combines  
the Job Title and its Risk Category.
```

```
# Function: paste0() joins strings without a space by  
default.
```

```
df_text <- ai_jobs_df %>%
```

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```
mutate(
  Job_Summary = paste0(Job_Title, " - ",
Risk_Category, " Risk")
)
print("--- Method C: Text Transformation (Job
Summary) ---")
print(head(df_text$Job_Summary))

#
=====
=====
# 5. ALL TOGETHER (The Standard Workflow)
#
=====
=====

final_dataset <- ai_jobs_df %>%
  mutate(
    # Calculation 1: Adjusted Salary (Salary scaled
by Tech Growth)
```

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```
Adjusted_Salary = Average_Salary *  
Tech_Growth_Factor,
```

```
# Calculation 2: Risk Flag (based on a  
combination of factors)
```

```
Is_High_Risk = ifelse(  
  Automation_Probability_2030 > 0.75 |  
Risk_Category == "High",  
  "Flagged",  
  "Safe"  
)
```

```
# Text Combo: A brief report on the job's risk  
factors
```

```
Risk_Report = paste0(  
  "AI Exposure: ", round(AI_Exposure_Index, 2),  
  " | Auto Prob: ",  
round(Automation_Probability_2030, 2),  
  " (" , Risk_Category, ")"  
)  
)
```

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```
print("--- Final Combined Dataset (Showing New  
Variables) ---")
```

```
print(head(final_dataset %>% select(Job_Title,  
Adjusted_Salary, Is_High_Risk, Risk_Report)))
```

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The screenshot shows the RStudio interface. The script editor contains R code for data analysis. The Environment pane on the right lists the objects created in the session.

```
# R Script: Creating New Variables (Transformations & Calculations)
# Dataset: AI Job Impact Data

library(dplyr)
library(tidyverse)
library(stringr)

# 1. SETUP: Import the Dataset

# Import data
ai_jobs_df <- read.csv("AI_Impact_on_Jobs_2030.csv")

print("---- Baseline Data Sample ----")
[1] "---- Baseline Data Sample ----"
# Show key columns used for the upcoming calculations
print(head(ai_jobs_df, 3) %>% select(Job_Title, Average_Salary, Years_Experience, AI_Exposure_Index))
# Job_Title Average_Salary Years_Experience AI_Exposure_Index
1 Security Guard 45795 28 0.18
2 Research Scientist 133355 20 0.62
3 Construction Worker 146216 2 0.86

# 2. METHOD A: ARITHMETIC CALCULATIONS
# Scenario 1: Calculate Salary Efficiency (Salary per Year of Experience).
# Formula: Average_Salary / Years_Experience

# Scenario 2: Create a composite "Future Readiness Score". (Higher score is better)
# Formula: (1 - Automation_Probability_2030) + (1 - AI_Exposure_Index) / 2

df_calc <- ai_jobs_df %>%
  mutate(
    Salary_Per_Year = Average_Salary / Years_Experience,
    Future_Readiness_Score = (1 - Automation_Probability_2030) + (1 - AI_Exposure_Index) / 2
  )

print("---- Method A: Arithmetic Results (Salary Efficiency & Readiness Score) ----")
[1] "---- Method A: Arithmetic Results (Salary Efficiency & Readiness Score) ----"
print(df_calc %>% select(Job_Title, Average_Salary, Years_Experience, Salary_Per_Year, Future_Readiness_Score))
# Job_Title Average_Salary Years_Experience Salary_Per_Year Future_Readiness_Score
1 Security Guard 45795 28 1635.536 0.485
2 Research Scientist 133355 20 6667.750 0.665
```

The Environment pane shows the following objects:

Object	Size	Modified
Final_dataset	3000 obs. of 21 variables	
Final_list	5 obs. of 3 variables	
Job_title	3000 obs. of 18 variables	
manual_split_df	5 obs. of 3 variables	
merged_data	5 obs. of 7 variables	
range_cols	3000 obs. of 3 variables	
selected_cols	3000 obs. of 3 variables	
split_list	List of 5	
split_matrix	chr [1:5, 1:2] "Security" "Research" "Construction" "Software" "Fin."	
starts_with_s	3000 obs. of 10 variables	
text_data	5 obs. of 2 variables	

The screenshot shows the RStudio console with the output of the R code. The output displays the results of the data analysis, including the calculated Salary_Per_Year and Future_Readiness_Score for each job title.

```
# 2. METHOD A: ARITHMETIC CALCULATIONS
# Scenario 1: Calculate Salary Efficiency (Salary per Year of Experience).
# Formula: Average_Salary / Years_Experience

# Scenario 2: Create a composite "Future Readiness Score". (Higher score is better)
# Formula: (1 - Automation_Probability_2030) + (1 - AI_Exposure_Index) / 2

df_calc <- ai_jobs_df %>%
  mutate(
    Salary_Per_Year = Average_Salary / Years_Experience,
    Future_Readiness_Score = (1 - Automation_Probability_2030) + (1 - AI_Exposure_Index) / 2
  )

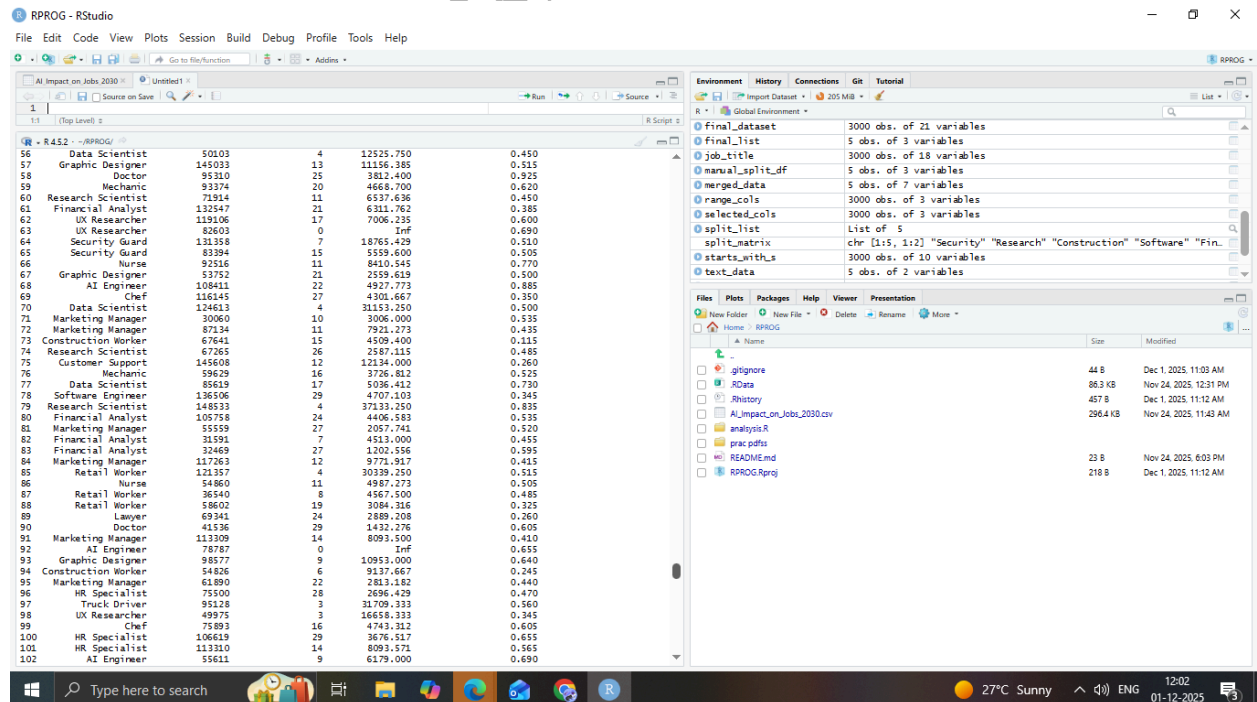
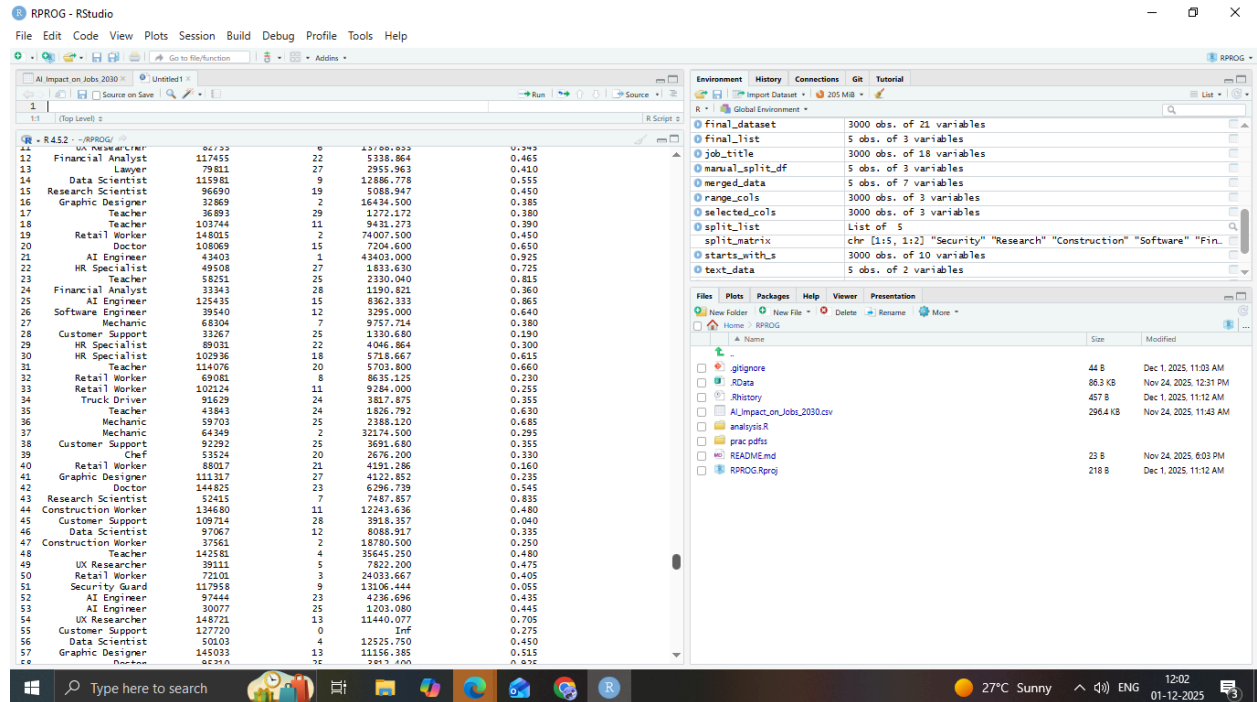
print("---- Method A: Arithmetic Results (Salary Efficiency & Readiness Score) ----")
[1] "---- Method A: Arithmetic Results (Salary Efficiency & Readiness Score) ----"
print(df_calc %>% select(Job_Title, Average_Salary, Years_Experience, Salary_Per_Year, Future_Readiness_Score))
# Job_Title Average_Salary Years_Experience Salary_Per_Year Future_Readiness_Score
1 Security Guard 45795 28 1635.536 0.485
2 Research Scientist 133355 20 6667.750 0.665
3 Construction Worker 146216 2 73108.000 0.165
4 Software Engineer 136530 13 10502.308 0.505
5 Financial Analyst 70397 22 3199.864 0.420
6 AI Engineer 92592 11 8417.455 0.805
7 Mechanic 107373 23 4668.391 0.460
8 Teacher 53419 12 4451.583 0.815
9 HR Specialist 139225 12 11602.083 0.610
10 Customer Support 85016 2 42508.000 0.595
11 UX Researcher 82733 6 13788.833 0.545
12 Financial Analyst 117455 22 5338.864 0.465
13 Lawyer 79111 27 2935.963 0.410
14 Data Scientist 115981 9 12886.778 0.555
15 Research Scientist 96690 19 5088.947 0.450
16 Graphic Designer 32869 2 16434.500 0.385
17 Teacher 36893 29 1272.172 0.380
18 Teacher 103744 11 9431.273 0.390
19 Retail Worker 148015 2 74007.500 0.450
20 Doctor 108069 15 7204.600 0.650
21 AI Engineer 43403 1 43403.000 0.925
22 HR Specialist 49508 27 1833.630 0.725
23 Teacher 58251 25 2330.040 0.815
24 Financial Analyst 33343 28 1190.821 0.360
```

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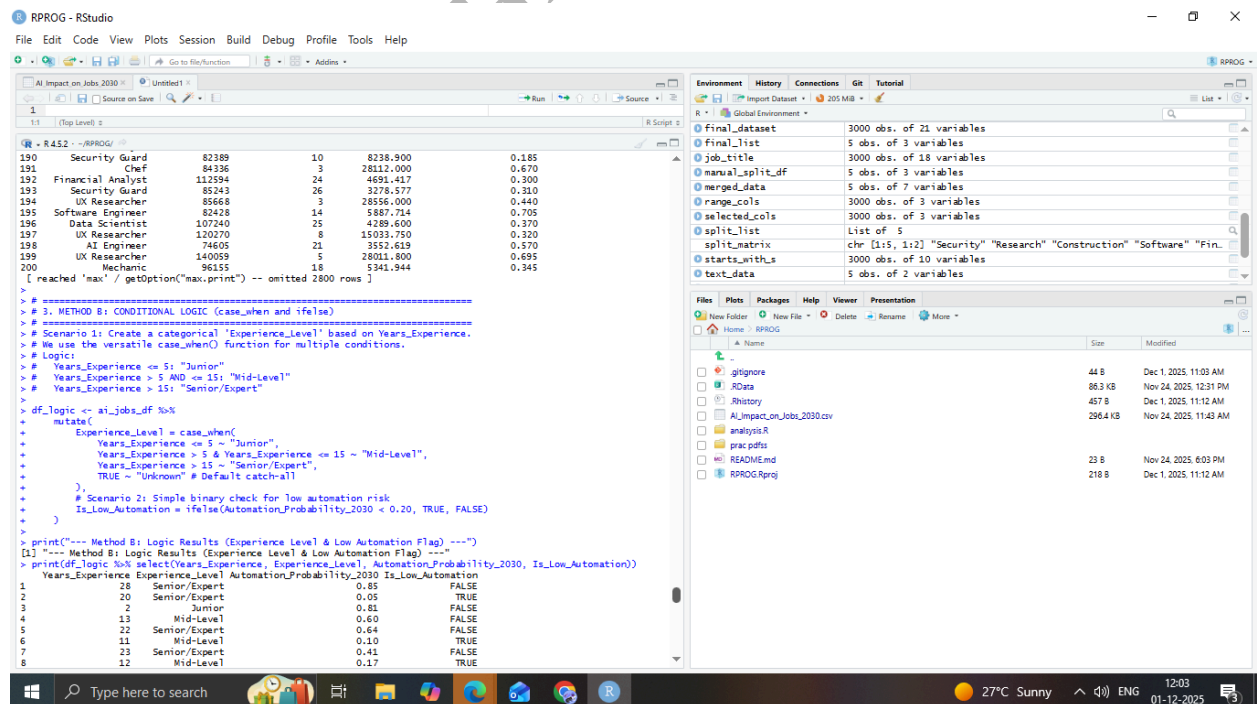
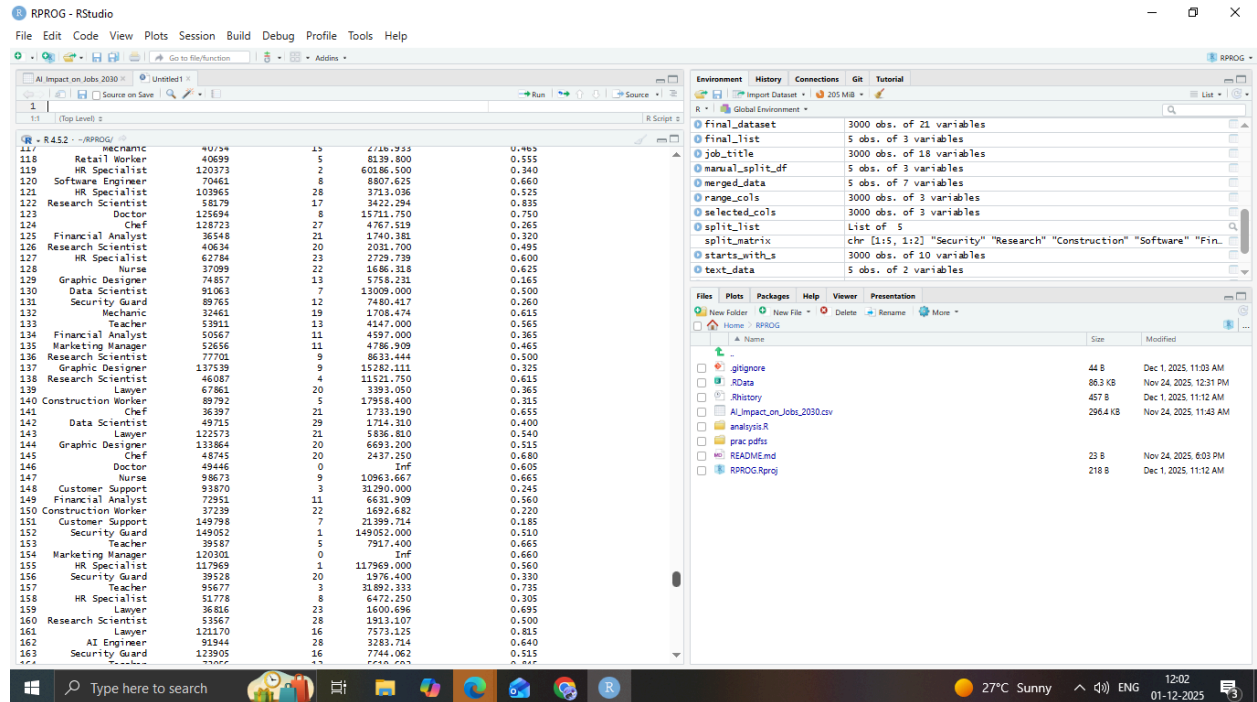


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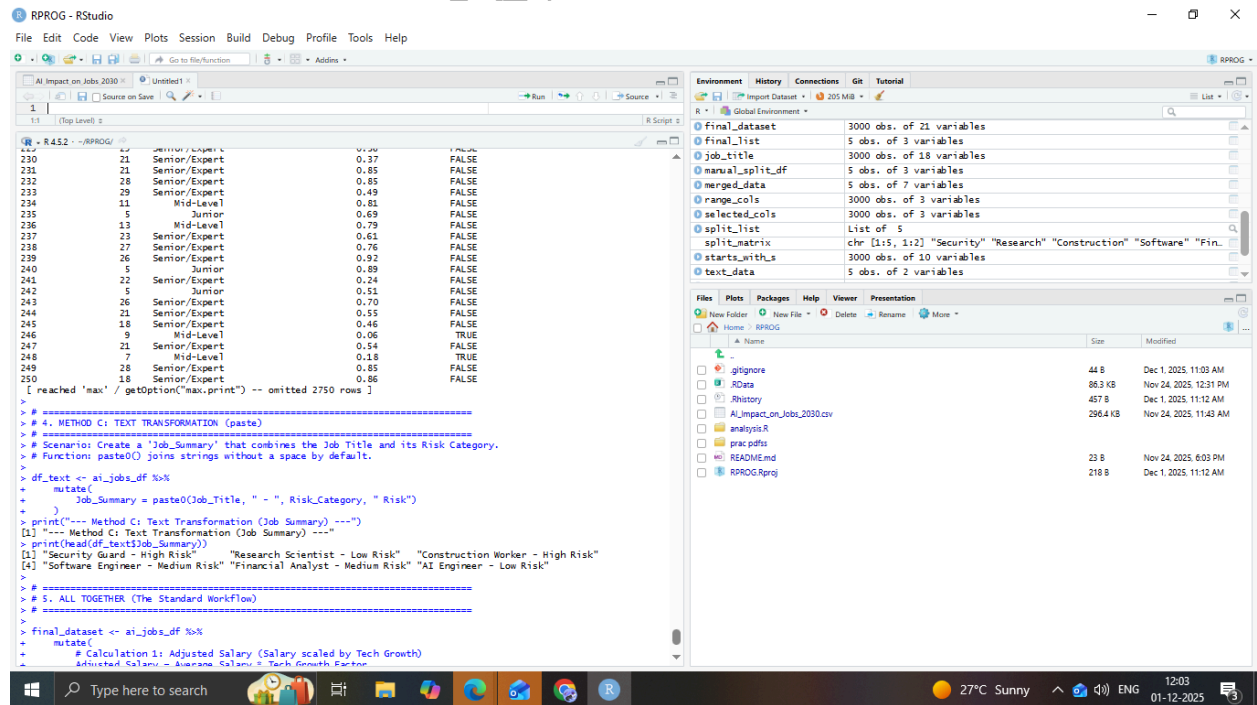
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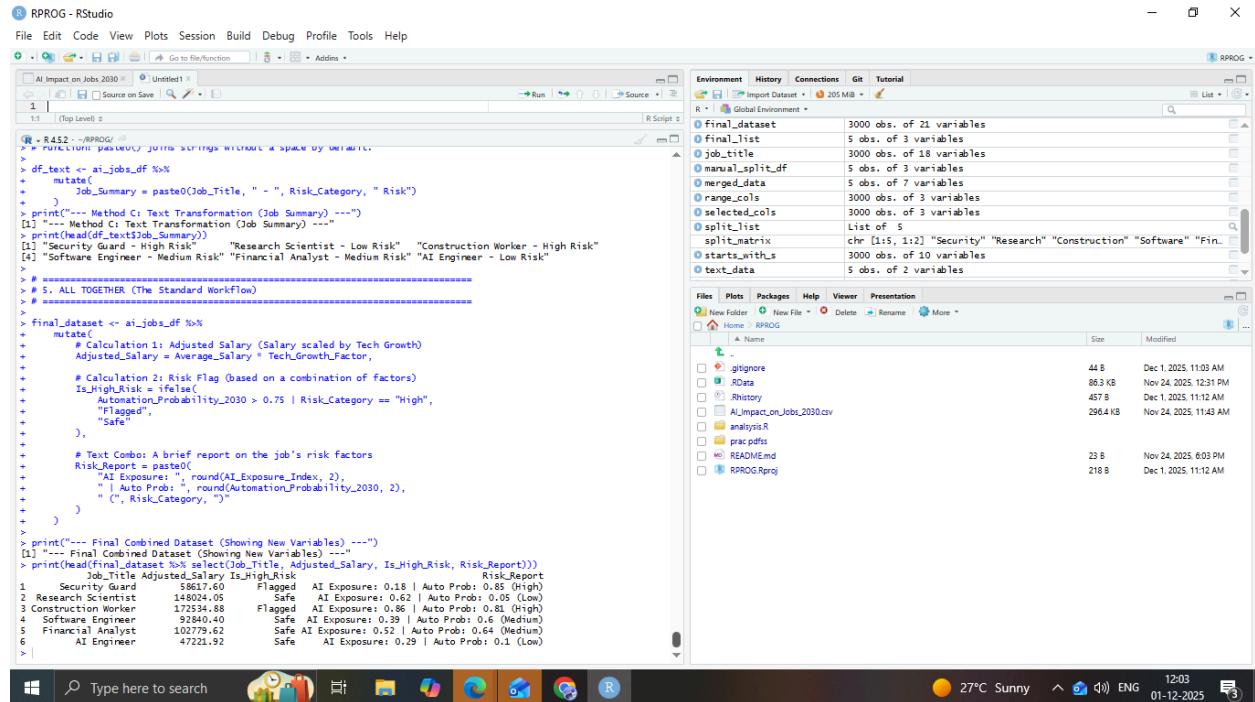
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The screenshot displays the RStudio environment. The main editor shows R code for data manipulation and analysis. The Environment pane on the right lists the objects created during the session.

```
R - R452 - RPROG
File Edit Code View Plots Session Build Debug Profile Tools Help
[+] AI Impact on Jobs 2030 [x] Untitled1
1 | (Top Level) 0

> # Functions: paste0() joins strings without a space by default.
> df_text <- ai_jobs_df %>%
+   mutate(
+     Job_Summary = paste0(Job_Title, " - ", Risk_Category, " Risk")
+   )
> print("---- Method C: Text Transformation (Job Summary) ----")
[1] "---- Method C: Text Transformation (Job Summary) ----"
> print(head(df_text$Job_Summary))
[1] "Security Guard - High Risk" "Research Scientist - Low Risk" "Construction Worker - High Risk"
[4] "Software Engineer - Medium Risk" "Financial Analyst - Medium Risk" "AI Engineer - Low Risk"

> # =====
> # 5. ALL TOGETHER (The Standard Workflow)
> # =====

> final_dataset <- ai_jobs_df %>%
+   mutate(
+     # Calculation 1: Adjusted Salary (Salary scaled by Tech Growth)
+     Adjusted_Salary = Average_Salary * Tech_Growth_Factor,
+     # Calculation 2: Risk Flag (based on a combination of factors)
+     Is_High_Risk = ifelse(
+       Automation_Probability_2030 > 0.75 | Risk_Category == "High",
+       "Flagged",
+       "Safe"
+     ),
+     # Text Combo: A brief report on the job's risk factors
+     Risk_Report = paste0(
+       "AI Exposure: ", round(AI_Exposure_Index, 2),
+       " | Auto Prob: ", round(Automation_Probability_2030, 2),
+       " (", Risk_Category, ")"
+     )
+   )
> print("---- Final Combined Dataset (Showing New Variables) ----")
[1] "---- Final Combined Dataset (Showing New Variables) ----"
> print(head(final_dataset %>% select(Job_Title, Adjusted_Salary, Is_High_Risk, Risk_Report)))
  Job_Title Adjusted_Salary Is_High_Risk Risk_Report
1 Security Guard      58617.60      Flagged AI Exposure: 0.18 | Auto Prob: 0.85 (High)
2 Research Scientist  148024.05       Safe  AI Exposure: 0.62 | Auto Prob: 0.05 (Low)
3 Construction Worker 172534.88      Flagged AI Exposure: 0.86 | Auto Prob: 0.85 (High)
4 Software Engineer   92840.40       Safe  AI Exposure: 0.39 | Auto Prob: 0.6 (Medium)
5 Financial Analyst   102779.62       Safe  AI Exposure: 0.52 | Auto Prob: 0.64 (Medium)
6 AI Engineer         47221.92       Safe  AI Exposure: 0.29 | Auto Prob: 0.1 (Low)

Environment History Connections Git Tutorial
R - Global Environment
[+] Import Dataset [x] 205 MB
[+] final_dataset 3000 obs. of 21 variables
[+] final_list 5 obs. of 3 variables
[+] job_title 3000 obs. of 18 variables
[+] manual_split_df 5 obs. of 3 variables
[+] merged_data 5 obs. of 7 variables
[+] range_cols 3000 obs. of 3 variables
[+] selected_cols 3000 obs. of 3 variables
[+] split_list List of 5
[+] split_matrix chr [15, 12] "Security" "Research" "Construction" "Software" "Fin..."
[+] starts_with 3000 obs. of 10 variables
[+] text_data 5 obs. of 2 variables

Files Plots Packages Help Viewer Presentation
[+] Home [x] RPROG
[+] Name Size Modified
[+] gignore 44 B Dec 1, 2025, 11:03 AM
[+] RData 86.3 KB Nov 24, 2025, 12:31 PM
[+] Rhistory 457 B Dec 1, 2025, 11:12 AM
[+] AI_Impact_on_Jobs_2030.csv 296.4 KB Nov 24, 2025, 11:43 AM
[+] analysis.R
[+] prac.pdfs
[+] README.md 23 B Nov 24, 2025, 6:03 PM
[+] RPROG.Rproj 218 B Dec 1, 2025, 11:12 AM
```

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